

What FSDS can attribute on one drifted offer across incentive, retention, and subsidy

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1 What is measured

One episode is one user. Three offers are scored and the top offer is kept. $Y = 1$ when that offer produces the scene outcome, and 0 otherwise. On the judge stream the habit offer is selected before user 16 and the click offer after it. Habit activates half of the users. Click activates multiples of five. So Y moves from 0.50 on the first 16 users to 0.167 on the next 24. That is eight fewer outcomes than the pre-drift rate. Each user scores three children either way. Search cost does not move.

X on the selected offer is (judge, stable, spurious, progress). On the judge stream the means move as follows.

Column	Pre-drift	Post-drift	Change
judge	0.820	0.808	-0.012
stable	0.500	0.200	-0.300
spurious	0.150	0.980	+0.830
progress	0.900	0.100	-0.800

The judge number barely moves. The path of the selection is the other three columns: the committed offer becomes the click offer, its spurious score jumps, and its progress score collapses. A permutation test of the mean shift separates every column from a shuffled past, because each column is a step at the same index. The size of the step is the interpretable part. The judge step is 0.012. The spurious step is 0.83.

2 What the concept test adds

Concept drift here is the drop in residual sum of squares from $Y \sim 1 + \text{batch}$ to $Y \sim 1 + \text{batch} + x + \text{batch} \cdot x$, with the batch split after the first 12 users. On the judge stream the four drops are 0.381, 0.381, 0.381, and 0.381, with permutation p -values 0.41, 0.41, 0.35, and 0.44. No column clears 0.10. The drops tie because, once the offer itself is a step at the drift, every column is a relabeling of that step. The interaction does not name a unique carrier of the fall in Y . The fall is carried by the batch bit: the offer changed. FSDS still returns a usable path, and it is the covariate path above, not a unique SSE ranking.

AUC is not an input. These episodes do not estimate an AUC, and they do not estimate a coefficient from AUC to conversion. A dollar figure built as $\Delta\text{AUC} \times \alpha \times V \times N$ is not a result of this run.

3 Two rounds of the spend rule

Unit prices below are assumptions used only to turn the measured counts into a ratio. $V = 1$ per outcome. Incentive cost 0.20 per treated user. Retention cost 0.50. Subsidy 0.40 on every paid user.

Round 1 spent when the running fit ranked spurious first. That rank appeared at user 19 and did not survive the full sample. Incentive then treated one later user, gained one outcome, spent 0.20, and finished at post-drift $Y = 0.208$, still below 0.50. Retention spent 1.0 and gained no outcome. Subsidy spent $0.40 \times 24 = 9.6$ because every user is paid, gained one order, and the ratio of extra orders to that spend is 0.15. The action text from the full-sample fit still says not to spend: the running rank and the full-sample rank disagreed.

Round 2 spends only when the top concept column is spurious, its SSE drop is positive, its mean has moved, and its concept p -value is at most 0.10. None of the four columns clears that bar. The lever stays off. Post-drift Y stays 0.167. Incentive and retention spend 0. Their ratio is undefined because the denominator is zero. Subsidy still pays 9.6, avoids 0 of the click waste, and has ratio 0 against the unmoved judge. Search cost remains three children per user.

4 The same path, read in three scenes

The measured path is one path. The scene changes which budget that path is allowed to open.

Scene	Reading of the same X move	Round-2 action	Money a
Incentive	Spurious up is not more participation.	Do not scale the incentive.	Spend 0.
Retention	Progress down is a worse offer, not a saved user.	Do not buy the save offer.	Spend 0.
Subsidy	Spurious up is the click offer taking the subsidy.	Do not cut the subsidy on this test.	Spend 9.

Incentive would become an increment only if a later action raised Y above 0.50. This fit does not authorize that action. Retention would be a partial recovery if the save offer were sent and Y moved back toward 0.50 without passing it. The concept test does not open that spend, so the shortfall is left in place. Subsidy already pays every user. The fraud reading is the spurious step of +0.83, and the concept p -values do not separate that column from the others, so the prototype does not convert the reading into a withheld subsidy.

5 Efficiency

The search scores the same three children before and after the drift, and before and after either round of the rule. Round 2 adds no outcome and no intervention. The information in the fit is the tied SSE drop plus the unequal mean shifts. Acting on the tied drop, as in round 1, buys at most one outcome and leaves every scene below its pre-drift rate. Waiting for a column that actually separates, as in round 2, spends nothing on incentive and retention and leaves the subsidy bill unchanged. That is the efficiency claim: the attribution is cheap to compute and expensive to obey too early.